MIDDLE EAST TECHNICAL UNIVERSITY
Department of Electrical and Electronics Engineering

EE 301 Signals & Systems
Homework 3

Due: November 19, 2017 (until 14:00 o'clock).

Please only submit the solutions for problems 3, 4, 5, 7 and 8. Notice that green-colored questions are different (at green-colored points) from their previous versions.

Problem 1: A continuous-time periodic signal $x_p(t)$ is real-valued and has a fundamental period $T_0 = 7$. The non-zero Fourier series coefficients of $x_p(t)$ are given as

$$a_0 = 1, \quad a_1 = a_{-1}^* = 1 + j, \quad a_3 = a_{-3}^* = -2$$

Express $x_p(t)$ in the form: $x_p(t) = \sum_{k=0}^{\infty} A_k \cos(\omega_k t + \varphi_k)$, that is, find the corresponding ω_k, A_k and φ_k values.

Problem 2: Let $x(t) = \begin{cases} 1 & 0 \le t < 1 \\ 2 - t & 1 \le t \le 2 \\ 0 & e.w. \end{cases}$

and $x_p(t)$ be a periodic extension of $x(t)$ such that $x_p(t) = \sum_{k=-\infty}^{\infty} x(t - kT)$ for $T = 4$.

$x_p(t)$ has the Fourier series coefficients a_k .

- **a)** Determine the value of a_0 .
- **b)** Determine the Fourier series representation of $dx_p(t)/dt$.
- **c)** Use the result of part (b) and the differentiation property of the Fourier series to help determine the Fourier series coefficients of $x_p(t)$.

Problem 3: Let $x_p(t)$ be a periodic signal whose Fourier series coefficients are

$$a_k = \begin{cases} 4, & k = 0 \\ [3^k - 3^{-k}], & otherwise \end{cases}$$

Use Fourier series properties to answer the following questions:

- **a)** Is $x_p(t)$ real?
- **b)** Is $x_p(t)$ odd or even?
- **c)** Is $dx_p(t)/dt$ odd or even?

Problem 4: Consider a causal CT and LTI system whose input $x(t)$ and output $y(t)$ are related by the following differential equation:

$$\frac{d}{dt}y(t) + 2y(t) = \frac{d}{dt}x(t)$$

Find the Fourier series representation of the output for each of the following inputs:

- **a)** $x(t) = \sin \pi t$
- **b)** $x(t) = \cos(2\pi t) + \sin(3\pi t + \frac{\pi}{6})$

Hint: You may substitute the FS expansion of a periodic input signal and a periodic output into the differential equation.

Problem 5: Suppose that a CT periodic signal is the input to an LTI system. The signal is given as

$$x_p(t) = \sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^k \cos\left(k\left(\frac{2\pi}{5}\right)t + \left(\frac{\pi}{6}\right)\right)$$

and the impulse response of the system satisfies

$$\int_{-\infty}^{\infty} h(t)e^{-j\omega t}dt = \begin{cases} 1, & |\omega| \le W \\ 0, & |\omega| > W \end{cases}$$

How large must the minimum value of W be in order for the output $y_p(t)$ of the system to have at least 90% of the average power per period of $x_p(t)$?

Hint: Try to use the Parseval's relation $\frac{1}{T_0}\int_{-\infty}^{\infty} |x_p(t)|^2 dt = \sum_{k=-\infty}^{\infty} |a_k|^2$ for the output signal $y_p(t)$.

Problem 6: Suppose we are given the following information about a signal $x(t)$:

1. $x(t)$ is a real signal.
2. $x(t)$ is periodic with $T = 4$ and has Fourier series coefficients a_k .
3. $a_k = 0$ for $k = 0$ and $k > 2$.
4. $x(t) = -x(t-2)$
5. $\int_{-2}^{2} |x(t)|^2\, dt = 2$
6. a_1 is a positive real number.

Show that $x(t)$ should be in the form: $Acos(Bt + C)$. Find $x(t)$.

Problem 7: Find the impulse response of the system with the frequency response

$$H(\omega) = \frac{e^{-6\omega j}}{3 + j\frac{\omega - 2}{5}}$$

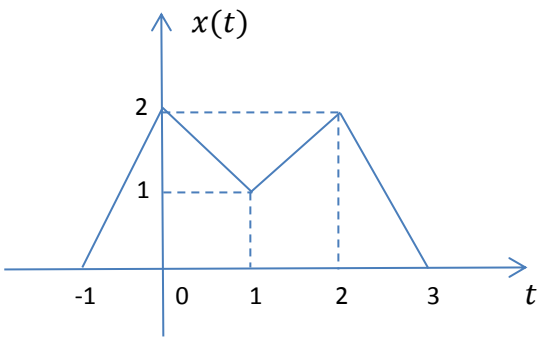
Problem 8: Let $X(\omega)$ denote the Fourier transform of the signal $x(t)$ depicted below. Perform the following calculations without explicitly evaluating $X(\omega)$.

a) $X(\omega)$ can be expressed as $A(\omega)e^{j\theta(\omega)}$ where $A(\omega)$ and $\theta(\omega)$ are both real-valued. Find $\theta(\omega)$.

b) Find $X(0)$.

c) Find $\int_{-\infty}^{\infty} X(\omega)\, d\omega$.

d) Sketch the inverse transform of $Re\{X(\omega)\}$.



(Notice that parts (d) and (e) of the previous version are skipped.)